

HCSM-33

The Mass Map

The Exponential Relation between the $N/4$ Anchors and
the Observed Particle Mass Spectrum

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation. The assignment of each mode to a specific Standard Model particle, together with the $N/4$ mass label attached to each, is the framework's *empirical thesis* (HCSM-23), structurally parallel to the Standard Model's Yukawa-coupling sector. The substrate derives the 14 modes, their coordinates, and the D_4 orbit structure; the $N/4$ labels are *empirical anchors*.

Given the anchor, the mass map $m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i)$ relates the $N/4$ anchors to the observed particle masses. The map's two coefficients have framework-constant forms: $A_{\text{mass}} = (2d-1)N/2 = 224$ and $B_{\text{mass}} = (d-1)kL^2/(2N^2) = 0.541197$, where $kL = \ln(M_P/v_{\text{EW}})$ is the electroweak Planck hierarchy (HCSM-51 [8]). Applied to the twelve charged particles (electron through top quark, plus W , Z , and Higgs), the map reproduces the observed masses at RMS 0.0449 dex approximately 11% accuracy spanning six orders of magnitude. The residual does not correlate with $N/4$ (Spearman $\rho = +0.414$, $p = 0.18$), consistent with noise.

The mass map is a *consistency check* on the framework's mass sector: it verifies that the anchored $N/4$ values and the framework constant coefficients A_{mass} and B_{mass} are jointly consistent with the observed masses at the 11% level. The map also has structural consequences: it predicts a neutrino mass $m_\nu = 1.77$ meV from the anchor $(N/4)_\nu = 60$, it relates the neutrino and Planck masses via the geometric mean form $m_\nu M_P = A_{\text{mass}} v_{\text{EW}} e^{2(d-1)}$ (used in HCSM-51), and it identifies the map rate B_{mass} with the DNLS interaction strength U (HCSM-18 [5]). All numerical claims are verified at machine precision.

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1 Introduction

1.1 The mass map in HCSM

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation [1, 3]. The assignment of each mode to a specific Standard Model particle, together with the $N/4$ mass label attached to each, is the framework’s *empirical thesis* (HCSM-23 [7]). It is not a derived result. The $N/4$ labels are empirical anchors, structurally parallel to the Standard Model’s Yukawa couplings.

The mass map relates the anchored $N/4$ values to the observed masses through the exponential relation

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i), \quad (1)$$

with two coefficients

$$A_{\text{mass}} = \frac{(2d-1)N}{2}, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2}. \quad (2)$$

The map top A_{mass} is a framework constant. The rate B_{mass} is a framework constant that depends on the warp factor $kL = \ln(M_P/v_{\text{EW}})$, which is derived in HCSM-51 [8] as $kL_{\text{bare}} = 38.4$ (matching the observed hierarchy 38.442527 to 0.11%).

Throughout this paper:

- The 14 modes, their coordinates, and the D_4 orbit structure are *derived* from the substrate [3].
- The $N/4$ mass label attached to each mode is an *empirical anchor* (HCSM-23 [7]).
- The coefficients A_{mass} and B_{mass} have framework constant forms. Their specific forms were identified by requiring the map to reproduce the observed mass spectrum; the origin of the forms is disclosed in §3.3.
- The map itself is a *consistency check* on the framework’s mass sector: it verifies that the anchored $N/4$ values and the framework constant coefficients are jointly consistent with the observed masses.

1.2 The one line summary

The mass map’s exponent spans from 0.27 (top) to 13.0 (electron) at the two ends of the anchor range. The exponential of this range covers the six orders of magnitude from the top mass to the electron mass. The map reproduces the twelve observed masses at RMS 0.0449 dex with framework-constant coefficients.

1.3 Derivation status of the results

The results of this paper split into three groups.

1.3.1 Framework-constant forms (input to the map)

The two coefficients of the mass map have framework-constant forms:

- (1) **Map top.** $A_{\text{mass}} = (2d-1)N/2 = 224$. The factor $(2d-1) = 7$ is a framework constant; the factor $N/2 = 32$ is a substrate-count constant. The specific combination $A_{\text{mass}} = 224$ was selected by the framework’s mass-fit test (§3.3).

- (2) **Map rate.** $B_{\text{mass}} = (d-1)kL^2/(2N^2) = 0.541197$ (at $d = 4$, $N = 64$, and observed $kL = 38.442527$). The form is a framework constant; the numerical value depends on the framework's warp factor kL (HCSM-51 [8]) and the observed hierarchy (which agrees to 0.11%).

1.3.2 The map's evaluation (empirical consistency check)

Given the anchor (the $N/4$ labels) and the framework constant coefficients, the map's evaluation against the observed masses is a *consistency check*:

- (3) **RMS accuracy.** The map reproduces the 12 particles at RMS 0.0449 dex (approximately 11%).
- (4) **Residual structure.** The residual does not correlate with $N/4$ ($\rho = +0.414$, $p = 0.18$), consistent with noise.
- (5) **Neutrino prediction.** The map gives $m_\nu = 1.77$ meV, consistent with the observed neutrino mass scale.

1.3.3 Structural consequences

The map also has structural consequences for the framework:

- (6) **Geometric-mean form.** The map is equivalent to the product identity $m_\nu M_P = A_{\text{mass}} v_{\text{EW}} e^{2(d-1)}$ (used in HCSM-51 to derive the closure condition).
- (7) **Relation to DNLS.** The map rate is identified with the DNLS interaction strength U (HCSM-18 [5]).

1.4 Cross references to the HCSM paper stack

The paper depends on the following companion papers:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate; the framework constants.
- HCSM-04 [3]: the 14-mode multiplet at $\lambda = 4$.
- HCSM-13 [4]: the processor T and the F/B -class correspondence.
- HCSM-18 [5]: the DNLS bridge (forward reference for the identification $B_{\text{mass}} = U$).
- HCSM-23 [7]: the identification thesis and the $N/4$ anchors.
- HCSM-51 [8]: the kL crossover.

1.5 Organization

Section 2 states the substrate, the 14 modes, and the $N/4$ anchors. Section 3 presents the mass map coefficients and the origin of their forms. Section 4 evaluates the map against the observed masses. Section 5 states the structural features of the map. Section 6 states the consistency with HCSM-51. Section 7 lists the falsifiability conditions. Section 8 catalogs inputs and derived. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup

2.1 Substrate and framework constants

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [2]. The framework constants are

$$d = 4, \quad N = d \cdot 2^d = 64, \quad L = \sqrt{N} = 8, \quad H = 2^d = 16, \quad V = (d - 1) \cdot 2^d = 48. \quad (3)$$

The discrete 5-point Laplacian has spectrum

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2, \quad (4)$$

with the self-paired eigenvalue $\lambda = 4$ of multiplicity 14.

2.2 The 14 modes and their coordinates

The 14 modes at $\lambda = 4$ decompose under the D_4 point group of the square into three orbits [3]:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (5)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (6)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad (7)$$

with sizes 2, 8, 4 and support classes 64, 48, 32, respectively.

2.3 The $N/4$ anchors

The $N/4$ values are the framework's empirical anchors for the particle spectrum (HCSM-23 [7]). In the standard mode order (matching the zero-state vector of HCSM-22 [6]):

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (8)$$

The values are treated as inputs throughout this paper. The structural relations among them are the theorems. Throughout, we use descriptive names for each mode (electron, muon, etc.), which correspond to the coordinate table of HCSM-23.

2.3.1 Relation to the WIN-era three-branch rules

The framework's WIN-era papers [9, 10] proposed three orbit-specific rules to reproduce the $N/4$ values from the mode coordinates: a 2^a power rule for $\mathcal{O}_2$, a fixed-pair rule for $\mathcal{O}_1$, and a cumulative-gap chain for $\mathcal{O}_0$. The rules reproduce the $N/4$ values (14/14 match) but contain free parameters — in particular the middle-orbit gap chain $\{1, 6, 0, 2, 2.5, 0.5, 1.5\}$ is not derived from first principles. In the HCSM framing, the $N/4$ values are treated as *empirical anchors*, not as outputs of the three-branch rules. The rules are a structural pattern, not a derivation.

2.4 The warp factor kL

The warp factor is $kL = \ln(M_P/v_{\text{EW}})$. HCSM-51 [8] derives the bare value $kL_{\text{bare}} = 38.4$ from the mass-map closure condition, matching the observed hierarchy $\ln(M_P/v_{\text{EW}}) = 38.442527$ to 0.11%. The observed value is 38.442527; the framework's bare value is 38.4.

For the numerical evaluation in this paper, we use the observed value $kL = 38.442527$. The framework-native value from HCSM-51 gives essentially the same result (§4).

2.5 The mass map form

The mass map is

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i), \quad (9)$$

with the coefficients (§3)

$$A_{\text{mass}} = 224, \quad B_{\text{mass}} = 0.541197. \quad (10)$$

The exponential form is not a free choice. The framework's prior attempts used a base-2 form $m = v_{\text{EW}} \cdot 2^{-N/4}$. That form reproduces only the top quark mass and fails by factors of 2 to 35 for the lighter particles. The exponential form $m = A \exp(-B \cdot N/4)$ is the correct functional form; the base-2 form is the special case $B = \ln 2 \approx 0.693$, which is excluded by the data (the fitted rate is 21% smaller).

3 The mass map coefficients

3.1 The map top A_{mass}

Definition 1 (Map top). The map top is

$$A_{\text{mass}} = \frac{(2d-1)N}{2}. \quad (11)$$

At $d = 4$, $N = 64$:

$$A_{\text{mass}} = \frac{7 \cdot 64}{2} = 224. \quad (12)$$

The expression factors as $A_{\text{mass}} = (2d-1) \cdot (N/2)$. The factor $(2d-1) = 7$ is a framework constant; the factor $N/2 = 32$ is half the substrate count. The composite is half the odd-mesh amplitude.

3.2 The map rate B_{mass}

Definition 2 (Map rate). The map rate is

$$B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2}. \quad (13)$$

At $d = 4$, $N = 64$, and $kL = 38.442527$:

$$B_{\text{mass}} = \frac{3 \cdot (38.442527)^2}{2 \cdot 64^2} = \frac{3 \cdot 1477.8276}{8192} = 0.541197. \quad (14)$$

The rate factors as $B_{\text{mass}} = \frac{d-1}{2} \cdot \frac{kL^2}{N^2}$. The factor $(d-1)/2 = 3/2$ is a framework constant; the factor kL^2/N^2 is the square of the warp factor normalized by the square of the substrate count.

With the framework's bare $kL = 38.4$ from HCSM-51:

$$B_{\text{mass}}^{\text{bare}} = \frac{3 \cdot (38.4)^2}{2 \cdot 64^2} = \frac{3 \cdot 1474.56}{8192} = 0.54. \quad (15)$$

The difference is 0.2%.

3.3 Origin of the coefficient forms

The specific forms of A_{mass} and B_{mass} were identified by requiring the mass map (1) to reproduce the observed mass spectrum. The WIN-era mass map paper searched a candidate set of 296,352 pairs (A, B) , each with a framework-constant form $A = \frac{p}{q} d^a N^b H^c V^e k L^f v_{\text{EW}}^g$ and similarly for B with small integer p, q and small integer exponents. The pair

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} \quad (16)$$

was selected as the form minimizing the RMS deviation from the twelve observed masses.

The forms are simple and have structural interpretations: $(2d-1)$ is the framework's odd-mesh coefficient, and $(d-1)/2$ is the framework's rigidity constant. Whether the specific combination selected by the search is derivable from first principles, or is one of several framework-constant forms that happen to fit well, is an open question in the framework. The present paper treats the forms as the framework's mass-sector constants, with the disclosure that their selection was made by fit.

Remark 3 (Relation to the SM Yukawa sector). The Standard Model's Yukawa couplings are 13 free parameters (one per charged fermion plus the Higgs quartic); the framework's mass map has two framework-expressed coefficients plus the anchor. The framework's claim is not that it has zero inputs, but that it has fewer than the Standard Model. The map's coefficients are analogous to the SM's Yukawa structure, with the anchor playing the role of the observed masses.

3.4 Relation to the DNLS interaction strength U

The mass map rate B_{mass} is identified with the DNLS interaction strength U on the 8×8 torus (HCSM-18 [5]):

$$U = B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.541197. \quad (17)$$

The identification is derived in HCSM-18 from the Lagrangian structure of the DNLS and the Boltzmann form of the mass map. In the mean-field approximation, the DNLS shifts the chemical potential by $\mu_\alpha = 4 + U \cdot \text{IPR}_\alpha$, giving three tiers separated by $U/128 = 0.0042281$.

The present paper takes the identification as an input from HCSM-18; the mass map itself does not depend on the DNLS derivation.

4 The map's evaluation

4.1 The 12 particles

Table 1 lists the map's predictions for the 12 particles with measured masses (electron through top quark, plus W , Z , and Higgs), alongside the observed values.

4.2 RMS accuracy

Theorem 4 (Mass map accuracy). *The mass map $m_i = 224 \exp(-0.541197 \cdot (N/4)_i)$ reproduces the 12 observed particle masses of Table 1 at*

$$\text{RMS} = \sqrt{\frac{1}{12} \sum_{i=1}^{12} (\log_{10} m_i - \log_{10} m_i^{\text{obs}})^2} = 0.0449 \text{ dex}. \quad (18)$$

Particle	$N/4$	m_{pred} (GeV)	m_{obs} (GeV)	Ratio	Residual (dex)
e	24.0	5.1206×10^{-4}	5.1100×10^{-4}	1.002	+0.0009
u	21.0	2.5968×10^{-3}	2.2000×10^{-3}	1.180	+0.0720
d	20.0	4.4614×10^{-3}	4.7000×10^{-3}	0.949	-0.0226
s	14.0	1.1474×10^{-1}	9.3000×10^{-2}	1.234	+0.0912
μ	14.0	1.1474×10^{-1}	1.0570×10^{-1}	1.085	+0.0356
c	9.5	1.3103	1.2700	1.032	+0.0136
τ	9.0	1.7175	1.7770	0.966	-0.0148
b	7.5	3.8677	4.1800	0.925	-0.0337
W	2.0	75.888	80.400	0.944	-0.0251
Z	2.0	75.888	91.200	0.832	-0.0798
H	1.0	130.38	125.25	1.041	+0.0174
t	0.5	170.89	172.70	0.990	-0.0046

Table 1: The mass map’s predictions vs. observed values. Residual is $\log_{10}(m_{\text{pred}}) - \log_{10}(m_{\text{obs}})$.

This corresponds to approximately 11% accuracy.

Proof. Direct computation from Table 1. The sum of squared residuals is 0.0241 dex²; dividing by 12 gives 0.00201; the square root is 0.0449. $\square$

The map’s accuracy with the framework-native $kL = 38.4$ from HCSM-51 (which gives $B_{\text{mass}} = 0.54$) is RMS 0.048 dex. The difference from the observed- kL value is 0.003 dex, negligible compared to the map’s 11% accuracy.

4.3 Residual structure

The residual does not correlate with $N/4$:

$$\text{Spearman}(\text{residual}, N/4) = +0.414, \quad p = 0.18. \quad (19)$$

The residual is consistent with noise at the 11% level. There is no systematic correction term suggested by the data.

The largest single residual is for the strange quark (+0.0912 dex), followed by the muon (+0.0356 dex) and the Z boson (-0.0798 dex). The W/Z pair shares the same $N/4 = 2$ anchor, so the map predicts the same mass for both; the observed W/Z mass difference (80.4 vs. 91.2 GeV) is a structural feature not captured by the map. This is one source of the map’s residual.

4.4 The neutrino

The map’s prediction for the neutrino, whose anchor is $(N/4)_{\nu} = 60$:

$$m_{\nu} = 224 \exp(-0.541197 \cdot 60) = 224 \cdot e^{-32.47} \approx 1.77 \times 10^{-12} \text{ GeV} = 1.77 \text{ meV}. \quad (20)$$

This is consistent with the observed neutrino mass scale (~ 1 meV) and with the framework’s previously-derived value $m_{\nu} = 1.7749$ meV from HCSM-51.

5 Structural features of the map

5.1 The exponential form

The map's exponent $B_{\text{mass}} \cdot (N/4)_i$ ranges from 0.271 at the top quark to 13.0 at the electron. The exponential of this range covers a factor of $e^{12.7} \approx 3.3 \times 10^5$; combined with the range of $N/4$ from 0.5 to 24 across the charged fermions, the map covers six orders of magnitude in mass.

The exponent is a Boltzmann-like activation factor. The framework's structural interpretation is that the mass of a particle is the equilibrium occupation probability of its mode in the vacuum configuration. This interpretation is developed in HCSM-18.

5.2 The geometric-mean identity

Theorem 5 (Geometric-mean form of the map). *The map applied to the neutrino, combined with the anchor $(N/4)_\nu = N - d = 60$, is equivalent to the product identity*

$$m_\nu \cdot M_P = A_{\text{mass}} \cdot v_{\text{EW}} \cdot e^{2(d-1)}. \quad (21)$$

At $d = 4$:

$$m_\nu \cdot M_P = 224 \cdot 246.22 \cdot e^6 \approx 2.17 \times 10^7 \text{ GeV}^2. \quad (22)$$

Proof. Immediate from the closure condition of HCSM-51 [8]: the mass map's depth at the neutrino is $B_{\text{mass}}(N - d) = kL - 2(d - 1)$, which rearranges to $m_\nu = A_{\text{mass}}(v_{\text{EW}}/M_P)e^{2(d-1)}$. Multiplying by M_P gives the stated identity. $\square$

The geometric-mean identity is the basis of HCSM-51's derivation of the warp factor kL . The map's self-consistency with the closure condition is what makes HCSM-51's derivation of kL work.

5.3 Relation to the three-tier DNLS structure

The DNLS mean-field on the 8×8 torus gives three tiers of chemical potential, separated by $U/128 = 0.0042281$, corresponding to the three D_4 orbits (sizes 2, 8, 4) and the three support classes (64, 48, 32). The mass map's coefficient $B_{\text{mass}} = U$ is the same quantity.

The mass map's residual structure across the three tiers is:

$$\text{Tier 1 (support 64, } \nu \text{ and } e\text{): RMS 0.0009 dex (electron only)} \quad (23)$$

$$\text{Tier 2 (support 48, 8 fermions): RMS 0.0512 dex} \quad (24)$$

$$\text{Tier 3 (support 32, } W, Z, H, t\text{): RMS 0.0423 dex} \quad (25)$$

The tier structure does not systematically bias the residual; the residual is approximately uniform across the three tiers.

6 Consistency with HCSM-51

HCSM-51 [8] derives the bare warp factor $kL_{\text{bare}} = 38.4$ from the mass-map closure condition

$$B_{\text{mass}} \cdot (N/4)_\nu = kL - \Delta, \quad \Delta = 2(d - 1) = 6. \quad (26)$$

The closure condition uses the map rate $B_{\text{mass}} = 0.541197$ and the neutrino anchor $(N/4)_\nu = 60$, both of which are inputs to the present paper.

The present paper's numerical evaluation uses the same coefficients: $A_{\text{mass}} = 224$ and $B_{\text{mass}} = (d - 1)kL^2/(2N^2)$ with $kL = 38.442527$. The consistency between the two papers is:

- HCSM-51 takes the map coefficients as inputs and derives kL .
- HCSM-33 takes the map coefficients and the anchor as inputs and evaluates the map against the observed masses.
- The kL value derived in HCSM-51 (38.4) and the observed value used here (38.442527) agree to 0.11%.

The two papers together close the mass-sector relations: HCSM-51 derives the warp factor from the mass-map closure condition; HCSM-33 evaluates the map against the observed masses. The closure is self-consistent to the map’s accuracy.

7 Falsifiability

The mass map is falsifiable by the following observations:

- (F1) **Wrong functional form.** If the observed mass spectrum required a functional form significantly different from $A \exp(-B \cdot N/4)$ (e.g., power-law, base-2 with a non-framework exponent, or with substantial cubic or higher-order corrections), the map’s structural form would be excluded.
- (F2) **Wrong coefficient forms.** If the observed masses were found to require coefficients substantially different from $A_{\text{mass}} = (2d - 1)N/2$ and $B_{\text{mass}} = (d - 1)kL^2/(2N^2)$ at fixed anchor values, the framework’s coefficient forms would be excluded.
- (F3) **Correlated residuals.** If the map’s residual were demonstrated to correlate significantly with a framework parameter (e.g., $N/4$, support class, or orbit), a systematic correction term would be required, invalidating the current map.
- (F4) **Neutrino mass outside 1–3 meV.** If the absolute neutrino mass were measured to lie outside the range suggested by the map ($m_\nu = 1.77$ meV), the map’s prediction would be falsified.
- (F5) **Anchored $N/4$ values inconsistent with observation.** If additional Standard Model particles were discovered with masses not fitting the anchor pattern of the 14 modes at $\lambda = 4$, the identification thesis (HCSM-23) would be falsified, and the map’s input would be invalidated.
- (F6) **W/Z mass difference not resolvable by higher-order structure.** The map predicts the same mass for W and Z (shared $N/4 = 2$); the observed mass difference (80.4 vs. 91.2 GeV) is a 13% residual. If this difference cannot be resolved by higher-order structure, the map’s structure at the $N/4$ -degenerate point would require modification.

The framework’s most specific prediction in the mass sector is the neutrino mass $m_\nu = 1.77$ meV. This is currently below the experimental sensitivity but will be testable by next-generation cosmological surveys and beta-decay experiments.

8 Inputs and derived

8.1 Derived from the substrate

The following are derived without reference to the $N/4$ anchors:

- (1) The framework constants $d = 4$, $N = 64$, $L = 8$, $H = 16$, $V = 48$ (HCSM-01 [2]).
- (2) The 14 modes at $\lambda = 4$ and their coordinates (HCSM-04 [3]).
- (3) The three D_4 orbits (sizes 2, 8, 4) and support classes (64, 48, 32).
- (4) The bare warp factor $kL_{\text{bare}} = 38.4$ (HCSM-51 [8]).
- (5) The identification $B_{\text{mass}} = U$ (HCSM-18 [5]).

8.2 Framework-constant forms (selected by fit)

- (6) The map top form $A_{\text{mass}} = (2d - 1)N/2 = 224$. The specific combination was selected from a candidate set of framework expressions by the mass-fit test (§3.3).
- (7) The map rate form $B_{\text{mass}} = (d - 1)kL^2/(2N^2) = 0.541197$. Same origin as (1).

8.3 Inputs

- (1) **Input 1 (SM content)**. The Standard Model particles and their observed masses, as used in the mass-fit test and the map’s evaluation.
- (2) **Input 2 (physical identifications)**. The identifications $A_1 \leftrightarrow \text{color}$, $B_1 \leftrightarrow \text{weak isospin}$.
- (3) **Input 3 (the identification thesis, HCSM-23)**. The assignment of each of the 14 modes at $\lambda = 4$ to a specific Standard Model particle, together with the $N/4$ mass label attached to each. This is the framework’s empirical anchor, structurally parallel to the Standard Model’s Yukawa couplings.
- (4) **The framework constants** $d = 4$, $N = 64$, $L = 8$, $H = 16$ (HCSM-01 [2]).

8.4 What is not claimed

We do not claim to derive the identification thesis (HCSM-23) from the substrate. The assignment of the 14 modes to the 14 particles, and the $N/4$ labels attached to each, are the framework’s empirical anchors.

We do not claim to derive the specific forms of the coefficients A_{mass} and B_{mass} from first principles. The forms have framework-constant expressions, and they were selected from a candidate set by the mass-fit test.

We do not claim that the mass map is a first-principles prediction of the particle spectrum. The map is a *consistency check* between the anchored $N/4$ values and the observed masses, given the framework-constant coefficients.

We do not claim to derive the EFT vacuum correction to the map rate; that is the subject of a companion paper.

The framework’s claim is that it has fewer empirical anchors than the Standard Model, not that it has none.

9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-33_verification`. It verifies:

- (1) The framework constants $d = 4$, $N = 64$, $L = 8$, $H = 16$, $V = 48$.
- (2) The coefficient values $A_{\text{mass}} = 224$ and $B_{\text{mass}} = 0.541197$.
- (3) The map's predictions for the 12 particles of Table 1.
- (4) The RMS accuracy 0.0449 dex.
- (5) The Spearman correlation $\rho = +0.414$, $p = 0.18$.
- (6) The neutrino prediction $m_\nu = 1.77$ meV.
- (7) The geometric-mean identity $m_\nu M_P = A_{\text{mass}} v_{\text{EW}} e^6$.
- (8) The map's RMS with the HCSM-51 value $kL = 38.4$ (which gives $B_{\text{mass}} = 0.54$): RMS 0.048 dex.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

10 Conclusion

This paper has established the mass map of the Hodge Complex Standard Model: the exponential relation

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i), \quad (27)$$

with framework-constant coefficients

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.541197, \quad (28)$$

and $N/4$ labels from the identification thesis (HCSM-23).

The map's coefficients have framework-constant forms. Their specific forms were selected by the framework's mass-fit test from a candidate set of framework expressions; the origin of the forms is disclosed but not derived.

Given the anchor and the coefficients, the map reproduces the twelve observed masses of the charged particles (electron through top quark, plus W , Z , and Higgs) at RMS 0.0449 dex (11% accuracy), spanning six orders of magnitude. The residual does not correlate with $N/4$, consistent with noise.

The map is a consistency check on the framework's mass sector: it verifies that the anchored $N/4$ values and the framework-constant coefficients are jointly consistent with the observed masses at the 11% level. The map also has structural consequences: it predicts a neutrino mass $m_\nu = 1.77$ meV, it is equivalent to the geometric-mean form used in HCSM-51 to derive the warp factor, and it identifies the map rate with the DNLS interaction strength.

The framework's claim is that the Standard Model is what the substrate looks like when you observe it. The identification of the 14 modes with the 14 particles is the empirical thesis; the mass map is the structural relation connecting the anchored $N/4$ labels to the observed masses. The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-13: The Processor T and the F/B -Class Correspondence*, HCSM White Paper Series (2026).
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- [6] S. Preschutti, *HCSM-22: The Zero State*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-23: The Identification of the 14 Modes with One Standard Model Generation and the $N/4$ Mass Labels*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-51: The kL Crossover*, HCSM White Paper Series (2026).
- [9] S. Preschutti, *The $5 + 9$ Split of the 14-Mode Multiplet*, WIN Paradigm White Paper (2026).
- [10] S. Preschutti, *A Zero-Parameter Derivation of the Particle Spectrum from the 8×8 Torus*, WIN Paradigm White Paper (2026).
- [11] Particle Data Group (S. Navas et al.), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024).

A Web Crawler Metadata

A.1 Dublin Core

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DC.subject: Hodge Complex Standard Model; HCSM; mass map; exponential mass relation; $N/4$ anchors; framework-constant coefficients; Warp factor; kL crossover; particle spectrum; identification thesis

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meV.

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A.2 BibTeX for Automated Indexing

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    Hodge Complex Standard Model relates the empirical N/4 anchors
    to the observed particle masses via the framework-constant
    coefficients  $A_{\text{mass}} = (2d-1)N/2 = 224$  and
 $B_{\text{mass}} = (d-1)k_L^2/(2N^2) = 0.541197$ . The map reproduces the
    12 charged particle masses at RMS 0.0449 dex ( $\sim 11\%$ ) and
    predicts  $m_{\nu} = 1.77$  meV.}
}
```